

4. The photographs show a pelican landing on water. The photographs are not in the correct order.



In which order were the photographs taken?

- (A) X, W, Z, Y (B) X, Z, W, Y (C) Y, Z, W, X (D) Y, Z, X, W
5. A man wanted to work out how tall he was by using the photograph that he took of himself standing next to a scale.



Using the scale, how tall is the man?

- (A) 190 cm
(B) 185 cm
(C) 170 cm
(D) 165 cm
6. Tess watched the Moon and wrote down what time it rose for five days in a row.

Monday	Tuesday	Wednesday	Thursday	Friday
6:01 pm	6:28 pm	6:57 pm	7:29 pm	8:05 pm

The information recorded in Tess' table shows that the moon

- (A) rose about 15 minutes later each day.
(B) rose about 30 minutes later each day.
(C) rose about 15 minutes earlier each day.
(D) rose about 30 minutes earlier each day.

1. The table shows the lengths of four types of dinosaurs.

Type of dinosaur	Length (m)
barosaurus	26
diplococus	30
supersaurus	34
turiasaurus	31

Which type of dinosaur was the longest?

- (A) barosaurus
(B) diplococus
(C) supersaurus
(D) turiasaurus

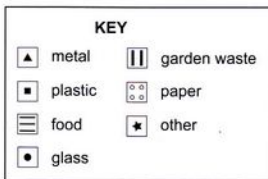
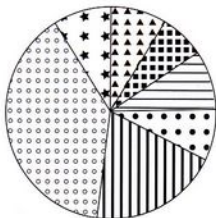
2. The photograph shows a slug beside a ruler. How long is the slug?



- (A) 5 cm (B) 4 cm (C) 3 cm (D) 1 cm

3. Most of what we throw away ends up as landfill.

This chart shows the different types of waste that can be found in landfill.

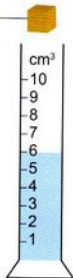


The largest amount of waste shown in this chart is

- (A) glass. (B) paper. (C) metal. (D) plastic.

7. Andy wanted to find the volume of a cube. His teacher told him that he could find this by placing the cube in water and measuring how much water it displaced. Volume can be measured in cubic centimetres (cm^3). He set up the experiment below.

block



step 1



step 2

What is the volume of the block, in cubic centimetres?

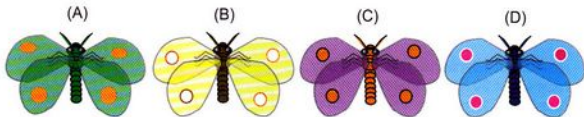
- (A) 1
(B) 2
(C) 6
(D) 8

8. Some birds eat butterflies. Butterflies that blend in with their surroundings are less likely to be eaten than those that do not.

The photograph shows a plant in a garden.



Which butterfly would be **least** likely to be eaten while resting on the leaves?



9. A group of students made a tally of the things they found in a sandpit.

leaves IIII IIII II

marbles III

rocks IIII IIII IIII I

sticks IIII (II)

Which option presents the correct data in the best way?

(A)

Number found	Things
leaves	12
marbles	3
rocks	16
sticks	8

(B)

Things	Number found
leaves	16
marbles	3
rocks	12
sticks	8

(C)

Number found	Things
12	leaves
3	marbles
16	rocks
8	sticks

(D)

Things	Number found
leaves	12
marbles	3
rocks	16
sticks	8

10. John wants to reduce the amount of sugar and sodium in his diet and increase the amount of protein.

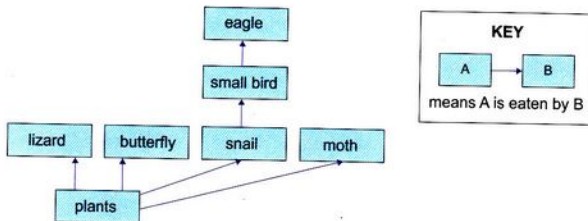
The table shows nutritional information for four foods in his diet.

	Chips	Lollies	Steak	Nuts
Protein (per 100 g)	7 g	0 g	26 g	27 g
Carbohydrate (per 100 g)	47 g	63 g	0 g	9 g
Sodium (per 100 g)	650 mg	110 mg	48 mg	500 mg
Sugar (per 100 g)	1 g	51 g	0 g	5 g

From the table, which food should John keep in his diet?

- (A) chips (B) lollies (C) steak (D) nuts

11. This is an example of a food web.



All of the small birds are killed off by a disease.

Which other group decreases in numbers as a result?

- (A) eagle (B) lizard (C) snail (D) moth

12. Blake studied the nests of some birds each spring.

He recorded the number of eggs he saw in one particular nest. He also recorded the condition of the nest.

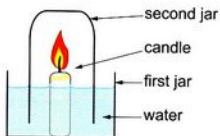
The table shows Blake's results.

Year	2005	2006	2007	2008	2009
Number of eggs	7	4	1	3	6
Condition of nest	very good	good	poor	good	very good

Which statement describes Blake's results?

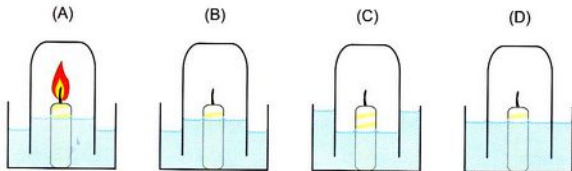
- (A) There were fewer eggs in the good nests than in the very good nests.
(B) Blake found more eggs in the poor nests than in the good nests.
(C) The bird did not lay any eggs in the poor nest.
(D) Both of the very good nests had seven eggs.

13. Sunil placed a candle in a jar of water. He then lit the candle and held a second jar over the top.



A short time later, the candle went out and water rose into the second jar.

Which diagram shows what happened?



14. Plants will grow best if their roots can grow down in soil that is loosely compacted.

Peter used a 15 cm long nail to measure how compacted the soil was in five different beds in his garden. In each bed, he pushed the nail into the soil until it became too hard to push further. Then he repeated this in a second location in the same bed. The table lists the depths the nail reached in the five different beds.

Garden bed	Depth location 1 (cm)	Depth location 2 (cm)	Average depth (cm)
bed 1	5	3	4
bed 2	4	6	5
bed 3	5	7	6
bed 4	3	3	3
bed 5	2	2	2

In which two garden beds were plants **least** likely to grow well?

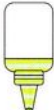
- (A) beds 1 and 4
 (B) beds 2 and 3
 (C) beds 3 and 5
 (D) beds 4 and 5

15. Detergent works by breaking up oil into very small droplets.

Frank tested four brands of dishwashing detergent. He filled four identical dropper bottles with different brands of detergent. For each test, he added drops of one brand of detergent to some oil, one drop at a time. He did this until all the oil had broken up into droplets.

Frank used a teaspoon of oil for each test.

The table shows the dropper bottles at the end of the test for the four brands of detergent.

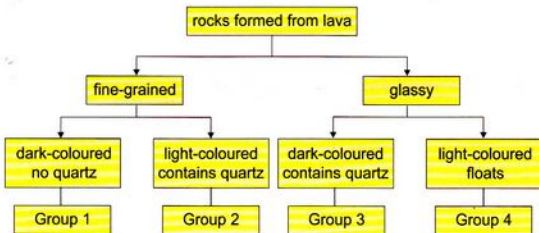
Test	1	2	3	4
Brand	Perfecto	Clean Up	Dish Lo	Sudsy
Dropper bottle at end of test				

Which brand of dishwashing detergent was the best at breaking up the oil?

- (A) Perfecto (B) Clean Up (C) Dish Lo (D) Sudsy

16. When a volcano erupts, lava flows from a vent and rocks are formed.

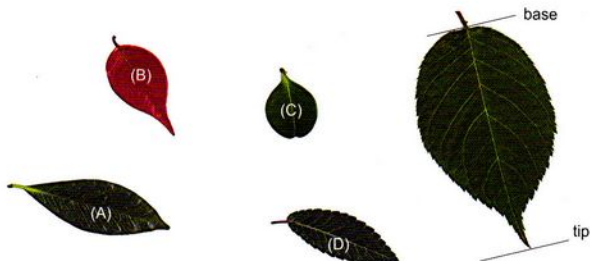
The diagram shows the different types of rocks that are formed.



What does a rock in Group 2 have in common with a rock in Group 4?

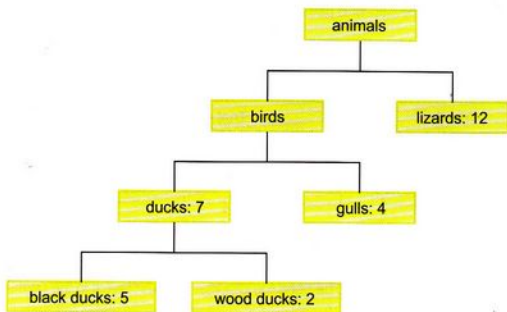
- (A) They are glassy. (B) They are fine-grained.
(C) They contain quartz. (D) They are light-coloured.

17. Marie's photograph shows leaves at their actual size. Which leaf measures 4 cm from base to tip?



18. Yusuf recorded the animals that lived near a pond. He put them into groups and counted how many in each group there were.

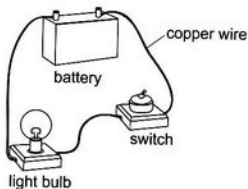
Here is his chart.



How many animals were **not** ducks?

- (A) 4 (B) 12 (C) 16 (D) 23

19. Rita and Helen built an electrical circuit.



They replaced the switch with different items: a coin, an eraser, a plastic key and a piece of wood.

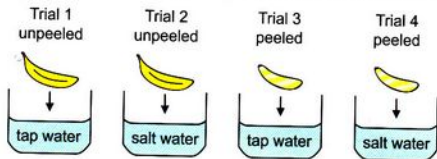
The light bulb lit up when they replaced the switch with the

- (A) coin. (B) eraser.
(C) plastic key. (D) piece of wood.

20. Mal wants to find the volume of a banana peel.

Mal dropped similar sized unpeeled and peeled bananas into containers of tap water and salt water.

He observed that unpeeled bananas floated just below the surface in tap water, but floated higher in salt water. The peeled bananas sank in both the tap water and the salt water.



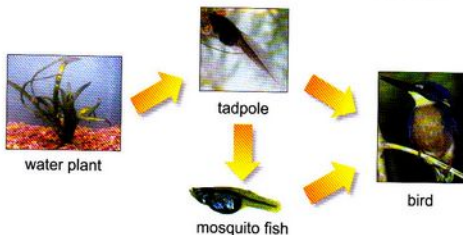
To work out the volume of a banana peel, he used the changes in water levels in the containers.

Which trials were the best ones for Mal to use?

- (A) Trials 1 and 2 (B) Trials 1 and 3
(C) Trials 2 and 3 (D) Trials 2 and 4

21. Food chains show the way energy moves as things are eaten.

The arrows in the food chain show what tadpoles eat and what eats them.



Which statement is correct about the tadpole food chain?

- (A) Mosquito fish eat water plants. (B) Tadpoles eat mosquito fish.
(C) Tadpoles eat birds. (D) Birds eat tadpoles and mosquito fish.

22. Peter wanted to find out if some materials were metals.

He used two rules:

- Metals must be shiny and not dull.
- Metals can be bent into a new shape and stay in that new shape.

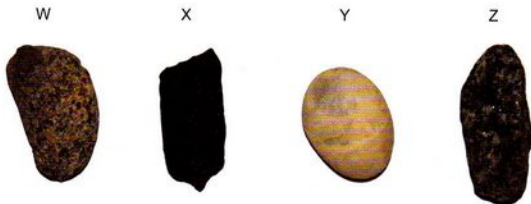
Here are Peter's results for four materials.

Material	Shiny	Bend and stay in new shape
1	no	no
2	yes	yes
3	yes	no
4	no	yes

Using the rules, which material is a metal?

- (A) material 1
(B) material 2
(C) material 3
(D) material 4

Ajay collected some rocks. The photograph below shows the actual size of four rocks.

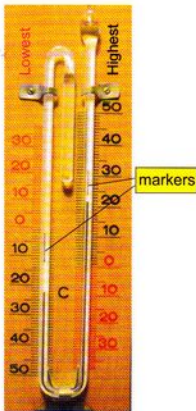


23. Which of the following is **not** an observation made using the photograph only?

- (A) Rock W is 2 cm wide.
- (B) Rock Y has the most egg-like shape.
- (C) Rocks W, X and Z contain some black particles.
- (D) Rock Z was rounded as it moved along a river bottom.

24. A special type of thermometer shows the lowest temperature and the highest temperature reached since it was last reset. The thermometer does this by moving the markers upwards. Then the markers stay in position until they are reset.

The thermometer is reset by moving the markers back down until they both are touching the silver mercury in the U-tube again.



What were the lowest and highest temperatures reached since this thermometer was last reset?

	Lowest temperature (°C)	Highest temperature (°C)
(A)	12	24
(B)	12	22
(C)	18	22
(D)	18	36

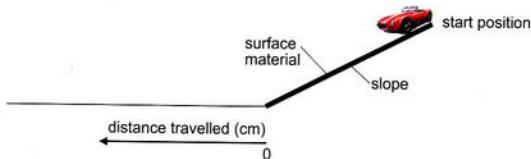
25. This photograph was taken while facing south on Australia's west coast.



When was the photograph taken?

- (A) in the morning
- (B) about midday
- (C) about midafternoon
- (D) very late in the afternoon

26. Georgie wanted to find the distance a toy car would travel after rolling down a slope.



She used three different surface materials on the slope. She rolled the car 3 times on each surface.

This table shows Georgie's results.

Surface	Distance travelled (cm)
cardboard	70, 85, 95
sandpaper	40, 65, 35
plastic	90, 110, 125

What can Georgie say about her results?

- (A) The surface of the slope did not make any difference to how far the car travelled.
- (B) The car always travelled further after the sandpaper slope than the plastic slope.
- (C) The car always travelled further after the plastic slope than the cardboard slope.
- (D) The car always travelled further after the cardboard slope than the sandpaper slope.

27. An insulator is a material that stops heat escaping.

Miriam did an experiment to investigate how insulators stop heat escaping from hot water. She used four glass jars of the same size and made lids from aluminium foil.

This is what she did:

- Step 1: She used a different number of layers of foil to make a lid for each jar.
- Step 2: She poured the same amount of hot water into each jar.
- Step 3: She measured the water temperature in each jar.
- Step 4: She fastened a lid to each jar with a rubber band.
- Step 5: She left the jars with their lids on for 30 minutes.
- Step 6: She measured the temperature of the water in each jar again.

What idea was Miriam investigating in this experiment?

- (A) Glass is good for keeping water warm.
- (B) Rubber bands are good for keeping water warm.
- (C) The amount of heat escaping depends on the number of layers of foil.
- (D) The amount of heat escaping depends on the amount of air in each jar.

28. Joyce poured the same amount of water into four different cylindrical jars.

She measured the surface area and the height of the water in each jar.

The table shows her results.

Jar	Surface area of water (cm^2)	Height of water (cm)
1	10	30
2	20	15
3	25	?
4	100	3

What is the height of the water in Jar 3?

- (A) 12 cm (B) 9 cm (C) 6 cm (D) 5 cm

29. A toy robot has four possible actions:

- turn head
- swing arms
- blink eyes
- move legs

Some of the actions do not work in faulty robots. The table shows the actions of six faulty robots.

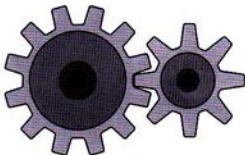
Faulty Robot Number	Action			
	turn head	swing arms	blink eyes	move legs
1	no	no	yes	no
2	no	yes	yes	yes
3	yes	no	no	no
4	yes	yes	no	yes
5	no	yes	no	yes
6	yes	no	yes	no

The four actions are controlled by three separate electrical circuits.

Which two actions are controlled by the same circuit?

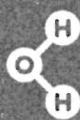
- (A) turn head and blink eyes (B) blink eyes and move legs
(C) turn head and swing arms (D) swing arms and move legs

30. Here are two gears.



If the large gear turned 40 times in a minute, how many times would the small gear turn in a minute?

- (A) 40 (B) 48 (C) 60 (D) 80



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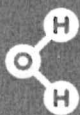
2010 Science Answer Keys

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Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Papers I & J
1	C	C	A	A	D	C	B	D	A
2	B	A	C	C	C	C	B	A	C
3	B	D	D	A	C	C	A	A	D
4	C	C	D	C	B	B	D	D	A
5	C	B	D	C	B	A	B	C	C
6	B	D	A	D	B	B	A	D	A
7	B	D	B	D	C	A	D	B	C
8	A	D	D	C	B	C	C	D	D
9	D	B	C	B	B	C	C	B	D
10	C	B	B	C	D	D	B	C	B
11	A	C	C	B	C	C	C	A	D
12	A	C	A	D	C	C	C	C	C
13	B	C	C	B	D	D	A	C	D
14	D	A	B	A	A	B	C	B	B
15	D	D	B	A	C	C	B	C	C
16	D	C	B	B	D	A	D	B	C
17	A	A	A	B	A	B	D	D	C
18	C	A	B	D	C	D	A	D	A
19	A	C	C	B	C	B	B	C	C
20	B	B	A	A	A	B	A	A	C
21	D	A	A	B	D	D	A	A	B
22	B	C	C	B	A	A	A	A	C
23	D	C	A	D	A	D	C	B	A
24	A	D	B	A	B	B	D	A	D
25	A	B	B	D	C	A	C	D	A
26	D	D	A	C	C	B	B	A	A
27	C	A	C	C	A	C	C	A	B
28	A	A	C	D	C	B	B	B	B
29	D	D	B	C	B	C	B	D	D
30	C	B	C	B	D	A	B	A	B

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31			C	B	B	B	B	D	A
32			B	C	B	D	D	A	A
33			B	D	B	B	B	C	D
34			D	B	D	B	D	B	D
35			A	D	B	D	C	B	C
36			D	D	A	D	B	C	C
37			B	A	A	D	D	A	B
38			A	B	C	C	A	D	A
39			D	A	B	D	D	B	D
40			C	A	D	D	D	C	C
41					A	A	B	C	B
42					D	A	A	B	A
43					D	A	B	B	B
44					D	B	C	C	B
45					B	B	C	C	A

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